Kuta Software - Infinite Precalculus Name____________________

# Power Series

Date________________ Period____

**Use the fifth partial sum of the exponential series to approximate each value. Round your answer to the nearest thousandth.**

1) $e^{0.4}$

2) $e^{-0.5}$

**Use the fifth partial sum of the power series for sine or cosine to approximate each value. Round your answer to the nearest thousandth.**

3) $\sin \frac{5\pi}{7}$

4) $\sin \frac{7\pi}{12}$

**Use $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ to find a power series representation of $g(x)$. Indicate the interval in which the series converges.**

5) $g(x) = \frac{3}{1-x^2}$

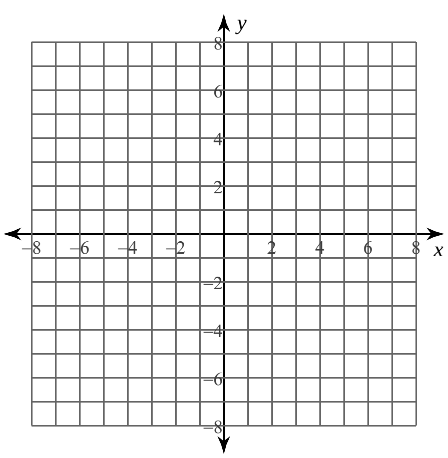
6) $g(x) = \frac{4}{3+2x^2}$

7) $g(x) = \frac{1}{1-x^2}$

8) $g(x) = \frac{3}{3-x^2}$

**Use $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ to find a power series representation of $g(x)$. Indicate the interval in which the series converges. Then graph $g(x)$ and the sixth partial sum of its power series.**

9) $g(x) = \frac{2}{2+x^2}$




10) $g(x) = \frac{1}{1-x^2}$